Online Math Olympiad 2013

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April 3, 2026

Selected solves from Evan Chen's Online Math Olympiad 2013.

Problem 12

Let a_n denote the remainder when $(n+1)^3$ is divided by n^3; in particular $a_1 = 0$. Compute the remainder when $a_1 + a_2 + \cdots + a_{2013}$ is divided by 1000.

Solution

The largest multiple of n^3 that is less than $(n+1)^3$ is

$$\left\lfloor \frac{(n+1)^3}{n^3} \right\rfloor n^3 = n^3 + \left\lfloor \frac{3n^2+3n+1}{n^3} \right\rfloor n^3.$$

Thus

$$a_n = (n+1)^3 - \left[n^3 + \left\lfloor \frac{3n^2+3n+1}{n^3} \right\rfloor n^3 \right].$$

It is easy to see that when $n \geq 4$ the fraction being floored is < 1 and so that whole term goes to 0. Hence from $n = 4, 5, \ldots, 2013$ we can telescope the sum like so

$$\sum_{n=4}^{2013} a_n = \sum_{n=4}^{2013} (n+1)^3 - n^3 = 2014^3 - 4^3.$$

Then it is easy to just compute $a_1 = 0, a_2 = 3, a_3 = 10$. So the problem reduces to evaluating

$$\begin{aligned} 2014^3 - 4^3 + 3 + 10 &\equiv 14^3 - 4^3 + 3 + 10 \pmod{1000} \\ &\equiv 2693 \equiv 693 \pmod{1000} \end{aligned}$$

So the answer is $\boxed{693}$.